Proposition 1. A subalgebra $\mathfrak{g} \subset \mathfrak{gl}(V)$ of nilpotent endomorphisms has a common eigenvector.

Proof. Induct on the dimension of the lie algebra. Dimension 1 is trivial, since for $0 \neq \xi \in \mathfrak{g}$, nilpotence of ξ implies all of its eigenvalues are zero and thus in k implying that ξ has a nonzero eigenvector.

Proceeding by induction let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a lie algebra of nilpotent endomorphisms having finite dimension greater than 1, and suppose the result holds for all lie algebras of dimension less that $\dim \mathfrak{g}$. Let $\mathfrak{h} \subset \mathfrak{g}$ be a nonzero proper subalgebra (one always exists since we can take the abelian subalgebra generated by a single element). Then we consider the vectorspace $U = \mathfrak{g}/\mathfrak{h}$, the adjoint representation of $\mathfrak{h}$ on U is nilpotent, thus by induction (since $\mathfrak{h} \subsetneq \mathfrak{g}$ has lesser dimension), $\mathfrak{h}$ acting on U has a common eigenvector. This shows that $\mathfrak{h}$ is properly contained in its normalizer $N_{\mathfrak{g}}(\mathfrak{h}) \supsetneq \mathfrak{h}$.

Now that we have established that for $\mathfrak{h} \subsetneq \mathfrak{g}$ a subalgebra that $N_{\mathfrak{g}}(\mathfrak{h}) \supsetneq \mathfrak{h}$ we can see that $\mathfrak{g}$ has a codimension 1 ideal. To see this, note that $N_{\mathfrak{g}}(\mathfrak{h})$ is a subalgebra of $\mathfrak{g}$ having greater dimension that $\mathfrak{h}$. Denote

$$N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) = \begin{cases} \mathfrak{h} & k = 0 \\ N_{\mathfrak{g}}(N_{\mathfrak{g}}^{(k-1)}(\mathfrak{h})) & k > 0 \end{cases} \tag{1}$$

by the previous argumentation, for $k > 0$ either $N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) = \mathfrak{g}$ or $\dim N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) > \dim N_{\mathfrak{g}}^{(k-1)}(\mathfrak{h})$, since each of these is a subalgebra of $\mathfrak{g}$ and $\dim \mathfrak{g}$ is finite the latter case can only happen finitely many times. Thus for some $k \geq 0$, we have $N_{\mathfrak{g}}^{(k)}(\mathfrak{h}) \subsetneq N_{\mathfrak{g}}^{(k+1)}(\mathfrak{h}) = \mathfrak{g}$. This establishes that $\mathfrak{g}$ has a nonzero proper ideal, which I will denote as I. To scaffold this into a codimension-1 ideal I study the adjoint representation on $\mathfrak{g}/I$. If $\ker \operatorname{ad} = \mathfrak{g}/I$, then $\mathfrak{g}/I$ is abelian, thus has a codimension 1 ideal and the preimage of this under the quotient map $\mathfrak{g} \to \mathfrak{g}/I$ is a codimension 1 ideal of $\mathfrak{g}$. If $\ker \operatorname{ad} = 0$, then by the first isomorphism theorem $\mathfrak{g}/I \cong \operatorname{ad}(\mathfrak{g}/I) \subset \mathfrak{gl}(\mathfrak{g}/I)$, then following the same argumentation used to produce I (∗), we find that $\operatorname{ad}(\mathfrak{g}/I)$ has a nonzero proper ideal J, and thus by the correspondence theorem, I is contained in a larger proper ideal $I \subsetneq J' \subsetneq \mathfrak{g}$, since $\mathfrak{g}$ is finite dimensional iterating this process we eventually get a codimension 1 ideal. Finally, if $0 \subsetneq \ker \operatorname{ad} \subsetneq \mathfrak{g}/I$, then by the correspondence theorem, $\ker \operatorname{ad}$ corresponds to a proper ideal J' with $I \subsetneq J' \subsetneq \mathfrak{g}$ so iterating this process we eventually get a codimension 1 ideal.

We have established existence of a codimension 1 ideal $\mathfrak{h} \subset \mathfrak{g}$. Applying the induction hypothesis, i.e. let $W \subset V$ be the nonzero subspace of eigenvectors for $\mathfrak{h}$, since $\mathfrak{h}$ is an ideal we have for $w \in W$, $\xi \in \mathfrak{g}, \eta \in \mathfrak{h}$

$$\eta\xi(w) = \xi \underbrace{\eta(w)}_{=0} + \underbrace{[\eta, \xi]}_{\in \mathfrak{h}}(w) = 0 \tag{2}$$

So that $\mathfrak{g} : W \to W$, now writing $\mathfrak{g} = \mathfrak{h} + k\xi$, we know the nilpotent endomorphism $\xi : W \to W$ has an eigenvector, this is already a common eigenvector for $\mathfrak{h}$ so it is a common eigenvector for $k\xi \oplus \mathfrak{h} = \mathfrak{g}$. □

(∗) To justify why the same argument works here, we first note that because every element of $\mathfrak{g}$ is a nilpotent operator, every element is ad-nilpotent (to see this simply compute the word $\operatorname{ad}_x^n(y)$), this of course implies ad-nilpotence for $\mathfrak{g}/I$, equivalently every element of $\operatorname{ad}(\mathfrak{g}/I)$ is a nilpotent operator in $\mathfrak{gl}(\mathfrak{g}/I)$. Moreover $\dim \mathfrak{g}/I < \dim \mathfrak{g}$ so it is valid to apply the induction hypothesis.

Theorem 1 (Engel)**.** If $\mathfrak{g} \subset \mathfrak{gl}(V)$ is a subalgebra of nilpotent operators, then there is a basis for V so that for every $x \in \mathfrak{g}$ the matrix representing x is strictly upper triangular.

Proof. It suffices to show that there exists a sequence of subspaces $V_j \subset V$ with $\dim V_j = j$ such that

$$0 = V_0 \subset V_1 \subset \cdots \subset V_{n-1} \subset V_n = V \tag{3}$$

and $\mathfrak{g}(V_j) \subset V_{j-1}$.

By **Proposition 1** $\mathfrak{g}$ has a common eigenvector $v \in V$, its eigenvalue must be zero since $\mathfrak{g}$ consists of nilpotent operators. Choose $V_1 = \operatorname{span}_k \{v\}$. Now assume that $V_1 \subset \cdots \subset V_r$ are vectorspaces with $\dim V_j = j$ and $r < n = \dim V$. Then $\mathfrak{g}$ acts on V/V_r, and since the action of any $x \in \mathfrak{g}$ on V is nilpotent, the same holds for the induced action on V/V_r. Thus we apply **Proposition 1** again to get a common eigenvector w with eigenvalue zero, let w' be a lift of w to V, then defining $V_{r+1} = V_r \oplus \operatorname{span} \{w'\}$ we are done by induction. □

Theorem 2 (Engel)**.** If every element of $\mathfrak{g}$ is ad-nilpotent, then $\mathfrak{g}$ is nilpotent.

Proof. By **Theorem 1** there is a basis for $\mathfrak{g}$ such that the matrix of every element of $\operatorname{ad}(\mathfrak{g})$ is strictly upper triangular. Since strictly upper triangular matrices in $\mathfrak{gl}(\mathfrak{g})$ are a nilpotent lie algebra so is their subalgebra $\operatorname{ad}(\mathfrak{g})$. This proves that $\operatorname{ad}(\mathfrak{g}) \cong \mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is nilpotent. To see this implies $\mathfrak{g}$ is nilpotent note that $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ being nilpotent implies that for some $n > 0$ we have the central series term $\mathfrak{g}^n \subset \mathfrak{z}(\mathfrak{g})$, then since $[\mathfrak{g}, \mathfrak{z}(\mathfrak{g})] = 0$ we find that the central series term $\mathfrak{g}^{n+1} = 0$ so that $\mathfrak{g}$ is nilpotent. □